

Divyansh Yadav

3rd Year Undergraduate, IIT Kanpur
Department of Mechanical Engineering

✉divyanshy23@iitk.ac.in | 📞 +91-9580436654
🌐 Divyansh Yadav | 🐙divyansh-iitk

Academic Qualifications			
Year	Degree/Certificate	Institute	CPI/%
2023 - Present	B.Tech	Indian Institute of Technology, Kanpur	6.8/10
2022	CBSE(XII)	Radiant Central Academy, Ambedkar Nagar	87.4%
2020	CBSE(X)	VSS Educational Academy, Ambedkar Nagar	87.8%

- Scholastic Achievements
- Secured **All India Rank 5592** in **JEE Advanced 2023** among the 2.5 lakh shortlisted candidates appearing for the examination
 - Secured **All India Rank 15354** in **JEE Mains 2023** among the 1.1 million registered candidates appearing for the examination

Key Projects

PDF ChatBot – RAG-based Document Q&A System

Self Project | 🐙

(May'26)

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline enabling users to upload PDF documents and ask natural language questions with accurate, context-grounded responses in real time
- Implemented document parsing using **PyMuPDF**, recursive chunking, **sentence-transformer** embeddings with **ChromaDB** and a hybrid retriever combining **BM25** keyword search with semantic similarity matching
- Developed a **FastAPI** backend with **LangChain** orchestration and **Groq**-hosted **Llama-3.3-70B** for context-aware response generation, along with a responsive **Streamlit** frontend supporting real-time chat and source attribution

GPT-2 from Scratch – Fine-tuned for Spam Classification

Self Project | 🐙

(Jul'25)

- Built a complete transformer-based **GPT-2 language model** from scratch in **PyTorch**, implementing core components including **multi-head self-attention**, **positional encoding**, **feed-forward layers** and **layer normalization** modules
- Loaded OpenAI's pre-trained GPT-2 weights into the custom architecture and fine-tuned the model for **spam classification**, achieving significantly improved accuracy and F1 score compared to traditional baseline methods on the benchmark dataset
- Applied **transfer learning** and **fine-tuning techniques** to effectively adapt the pre-trained model for solving a range of downstream NLP tasks

AES-128 Implementation and Side-Channel Attack (CPA) | 🐙

Course Project | CS698G | Prof. Udaya Parampalli

(May'25-Jul'25)

- Implemented the complete **AES-128 encryption algorithm** from scratch in Python, building all core transformation stages including **key expansion**, **SubBytes**, **ShiftRows**, **MixColumns** and **AddRoundKey** operations
- Designed and executed a full **Correlation Power Analysis (CPA)** side-channel attack targeting the first encryption round, analyzing simulated power consumption traces to successfully recover all individual cryptographic key bytes
- Demonstrated that even mathematically strong and theoretically secure cryptographic systems can be compromised through practical **side-channel vulnerabilities** such as power analysis, timing attacks and electromagnetic emanation leaks

- Technical Skills
- Programming Languages:** Python, C, C++, Bash, $\text{\LaTeX}$
 - Libraries & Frameworks:** PyTorch, LangChain, FastAPI, Streamlit, Scikit-learn, Pandas, NumPy, Matplotlib, Sentence-Transformers, spaCy
 - Tools & Platforms:** Git, GitHub, ChromaDB, Jupyter

Fundamentals of Data Science and Machine Intelligence Applied Cryptography Linear Algebra Single Variable Calculus Introduction to Electronics	Software Development and Operations Fundamentals of Computing Application of Single Variable and Multivariable Calculus Differential Equations Robotics
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Extra-Curricular Activities

GDSC Diagonocode Hackathon Winner

(Mar'24)

- Built an **NLP-based** system during a **GDSC** hackathon to automatically identify and extract drug names from noisy user reviews using classical machine learning and text processing techniques
- Used **spaCy** for part-of-speech tagging to extract candidate nouns, and **difflib** for fuzzy string matching to mask and select the most likely drug names from the extracted candidates
- Designed a **minimalist** and resource-efficient solution without relying on heavy transformer models, keeping the approach **simple**, **lightweight** and **effective** for rapid deployment